In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE (Advanced), 2017

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - IV

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 186**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 05)** contains 5 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (06 to 13) contains 8 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer and **– 2 mark** for wrong answer.

(ii) **Section-C (01 to 05)** contains 5 Numerical based questions with single digit integer as answer, ranging from 0 to 9. Each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

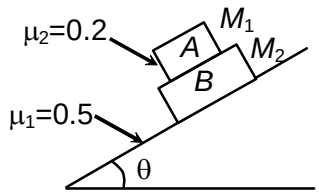
SECTION – A

Straight Objective Type

This section contains **5 multiple choice questions** numbered 1 to 5. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. Two trains *A* and *B* are moving on same track in opposite direction with velocity 25 m/s and 15m/s respectively. When separation between them becomes 225 m, drivers of both the trains apply brakes producing uniform retardation in train *A* while retardation of train *B* increases linearly with time at the rate of 0.3 m/s^3. The minimum retardation of train *A* to avoid collision will be
(A) 2 m/s^2 (B) 2.5 m/s^2
(C) 2.25 m/s^2 (D) 2.75 m/s^2

2. Two blocks M_1 and M_2 rest upon each other on an inclined plane. Coefficient of friction between surfaces are shown. If the angle θ is slowly increased, then ($M_1 < M_2$)
(A) block *A* slips first
(B) block *B* slips first
(C) both slip simultaneously
(D) both remain at rest



3. An artificial satellite of mass *m* is moving in a circular orbit at a height equal to the radius *R* of the earth. Suddenly due to internal explosion the satellite breaks into two parts of equal masses. One part of the satellite stops just after the explosion. The increase in the mechanical energy of the system (satellite + earth) due to explosion will be
(Given: acceleration due to gravity on the surface of earth is *g*)
(A) mgR (B) $\frac{mgR}{2}$
(C) $\frac{mgR}{4}$ (D) $\frac{3mgR}{4}$

Space for rough work

4. The equation of a particle executing SHM is given by $x = 3\cos\left(\frac{\pi}{2}\right)t$ cm, where t is in second. The distance travelled by the particle in the first 8.5 s is

(A) $\left(24 + \frac{3}{\sqrt{2}}\right)$ cm (B) $\left(27 - \frac{3}{\sqrt{2}}\right)$ cm

(C) $\left(24 - \frac{3}{\sqrt{2}}\right)$ cm (D) $\left(27 + \frac{3}{\sqrt{2}}\right)$ cm

5. n identical charged particles are placed on the vertices of a regular polygon of n sides of side length a. One of the charge particle is released from polygon. When this particle reaches a far of distance, another particle adjacent to the first particle is released. The difference of kinetic energies of both the particles at infinity is k. Magnitude of charge is

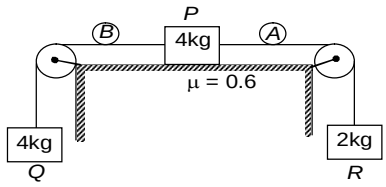
(A) $\sqrt{4\pi\varepsilon_0 ak}$ (B) $\frac{k}{4\pi\varepsilon_0 a}$

(C) $\frac{k}{a}$ (D) $\sqrt{ka}$

Multiple Correct Choice Type

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

6. A blocks P of mass 4 kg is placed on horizontal rough surface with coefficient of friction μ = 0.6. And two blocks R and Q of masses 2 kg and 4 kg connected with the help of massless strings A and B respectively passing over frictionless pulleys as shown, then (g = 10m/s^2)



(A) acceleration of block P is zero.
(B) tension in string A is 20 N.
(C) tension in string B is 40 N.
(D) surface contact force on block P is $20\sqrt{5}$ N.

7. A body moving towards a finite body at rest collides with it. It is possible that
(A) both the bodies come to rest
(B) both the bodies move after collision
(C) the moving body comes to rest and the stationary body starts moving
(D) the stationary body remains stationary; the moving body changes its velocity

Space for rough work

8. An object is floating in a liquid, kept in a container. The container is placed in a lift. Choose the correct option(s)
(A) Buoyant force increases as lift accelerates up
(B) Buoyant force decreases as lift accelerates up
(C) Buoyant force remains constant as lift accelerates
(D) The fraction of solid submerged into liquid does not change

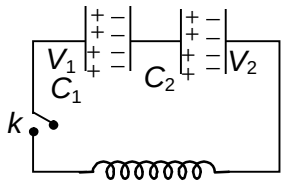
9. A standing wave of time period T is set up in a string clamped between two rigid supports. At $t = 0$ antinode is at its maximum displacement $2A$.
(A) The energy of a node is equal to energy of an antinode for the first time at $t = T/8$.
(B) The energy of node and antinode becomes equal after every $T/2$ second.
(C) The displacement of the particle of antinode at $t = \frac{T}{8}$ is $\sqrt{2}A$.
(D) The displacement of the particle of node is zero.

10. Charges Q_1 and Q_2 lie inside and outside respectively of a closed Gaussian surface S. Let E be the field at any point on S and ϕ be the flux of E over S:
(A) If Q_1 changes, both E and ϕ will change
(B) If Q_2 changes E will change
(C) If $Q_1 = 0$ and $Q_2 \neq 0$ then $E \neq 0$ but $\phi = 0$
(D) If $Q_1 \neq 0$ and $Q_2 = 0$ then $E = 0$ but $\phi \neq 0$

11. Two circular coils of radii 5 cm and 10 cm carry equal currents of 2A. The coils have 50 and 100 turns respectively and are placed in such a way that their planes as well as their centres coincide. Magnitude of magnetic field at the common centre of coils is
(A) $8\pi \times 10^{-4}$ T if current in the coil are in same sense
(B) $4\pi \times 10^{-4}$ T if current in the coil are in opposite sense
(C) zero if currents in the coils are in opposite sense
(D) $8\pi \times 10^{-4}$ T if current in the coil are in opposite sense

Space for rough work

12. Two capacitors of capacitance C_1 and C_2 are charged to a potential difference of V_1 and V_2 respectively and are connected to an inductor of inductance L as shown in the figure. Initially key k is open. Now key k is closed and current in the circuit starts increasing. When current in the circuit is maximum



(A) Charge on both the capacitors is same
(B) Induced emf in the inductor is zero
(C) Potential difference across both the capacitors is same
(D) Electrostatic potential energy stored in both the capacitors is same

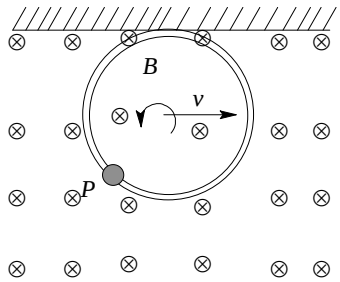
13. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy T_A eV and de Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.70 eV is $T_B = (T_A – 1.50)$eV. If the de Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$, then
(A) The work function of A is 1.25 eV (B) the work function of B is 4.20 eV
(C) T_A = 3.00 eV (D) T_B = 1.5 eV

SECTION – C

(One Integer Value Correct Type)

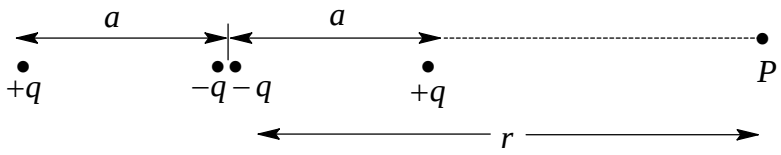
This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. A rigid ring is made to roll along the ceiling of a room, where exists a uniform horizontal magnetic field of induction B perpendicular to the plane of the ring. The velocity of the centre of the ring is constant and its modulus is v. A charge particle P of mass m is fixed on the ring. The charge q on the particle is $\frac{kmg}{Bv}$ so that there is no force of interaction between the ring and the particle, then find the value of k ?

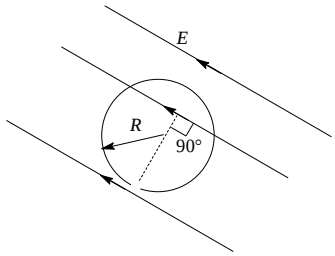


Space for rough work

2. Figure shows a charge array known as an "electric quadrupole". For a point on the axis of the quadruple, the dependence of potential on r $\left(\frac{r}{a} >> 1\right)$ is $\frac{1}{r^n}$. Find n ?



3. A non conducting ring having uniform charge distribution q is having a small cut of length $b\left(b << R\right)$ is released from rest with its plane parallel to the field as shown in the figure. If the maximum angular velocity of the ring is $\sqrt{\frac{C\,qEb}{22mR^2}}$, then find the value of C ?



4. Two wires are made of the same material and have the same volume. However, wire 1 has cross-sectional area A and wire-2 has cross-sectional area 3A. If the length of wire 1 increases by Δx on applying force 1 newton, how much force (in Newton) is needed to stretch wire 2 by the same amount?

5. Two pendulums of lengths 1.44 and 1 metre length start swinging together. After how many oscillation of first pendulum they will again start swing together?

Space for rough work

Chemistry

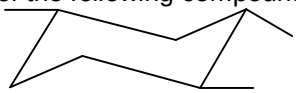
PART – II

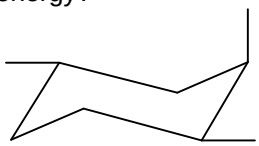
SECTION – A

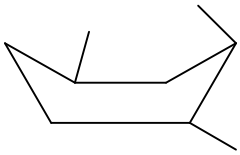
Straight Objective Type

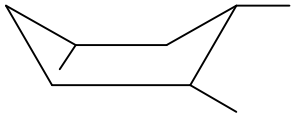
This section contains **5 multiple choice questions** numbered 1 to 5. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. Which of the following compound has lowest potential energy?

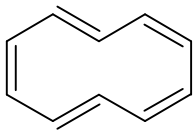
(A) 

(B) 

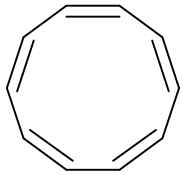
(C) 

(D) 

2. The following two compound are:



(I)



(II)

(A) Identical
(B) conformational isomers
(C) geometrical isomers
(D) structural isomers

3. If mercuric iodide is added to an aqueous solution of KI, the:
(A) freezing point is raised
(B) freezing point is lowered
(C) boiling point does not change
(D) freezing point does not change

Space for rough work

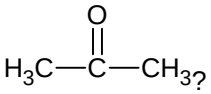
4. Which of the following is weaker base?
(A) $N\left(SiH_3\right)_3$ (B) $N\left(CH_3\right)_3$
(C) $NH\left(CH_3\right)_3$ (D) All are equally basic

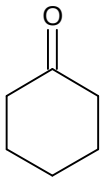
5. Choose the correct method for each:
(A) Ni : electrolysis, Cu : van Arkel process, Zn : zone refinining, Ga : zone refining
(B) Ni : mond's process, Cu : electrolysis, Zr : van Arkel process, Ga : zone refining
(C) Ni : electrolysis, Cu : zone refining, Zr : van Arkel process, Ga : electrolysis
(D) Ni : electrolysis, Cu : zone refining, Zr : van Arkel process, Ga : mond's process

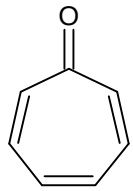
Multiple Correct Choice Type

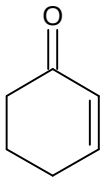
This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

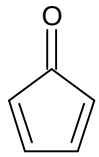
6. Which of the following compound have larger $C-O$ bond length than $C-O$ bond length of



(A) 

(B) 

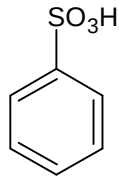
(C) 

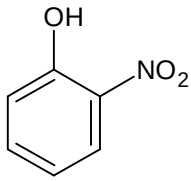
(D) 

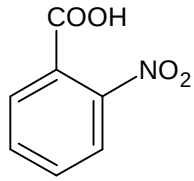
Space for rough work

7. Which of the following compounds are soluble in $NaHCO_3$?

(A) Picric acid

(B) 

(C) 

(D) 

8. Which of the following statement/statements is/are correct?

(A) in the $NaCl$ structures, each Na^+ ion is surrounded by $6Cl^-$ ions as nearest neighbours and twelve Na^+ ions as next nearest neighbours.

(B) the co-ordination number of cation generally increases as the radius ratio increases

(C) packing efficiencies higher than 0.74 are possible only with cation and anion of different sizes

(D) during a crystal defect electrical neutrality should be maintained.

9. Which of the following has/have a dipole moment:

(A) $2,2,3,3-$ tetramethylbutane

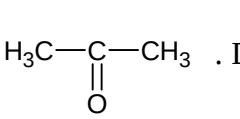
(B) Trans-2-pentene

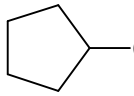
(C) cis-1, 2-dichloroethene

(D) trans-1, 2-dichloroethene

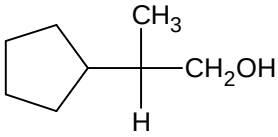
Space for rough work

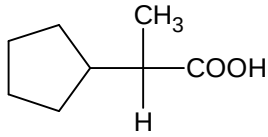
10. An optically active alcohol $A(C_8H_{16}O)$ on oxidation gives B. A on acidic heating gives $C(C_8H_{14})$ as major product. C on ozonolysis produces $D(C_5H_8O)$ and

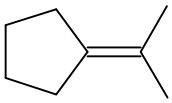


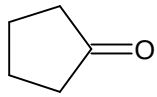
. D on reduction with $LiAlH_4$ gave 

Identify correct answer:

(A) A is 

(B) B is 

(C) C is 

(D) D is 

11. A mixture of three proteins, (A) (pepsin), (B) (haemoglobin) and (C) (Lysozyme) was separated by electrophoresis method at pH=7. The pH at isoelectric point (pI) of the proteins are pI of (A), (B) and (C) which are following statement/statements is/are correct?

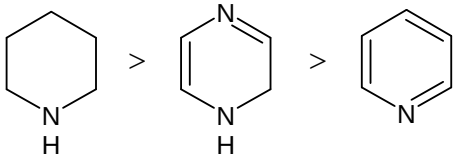
(A) pepsin (A) will migrate to the cathode

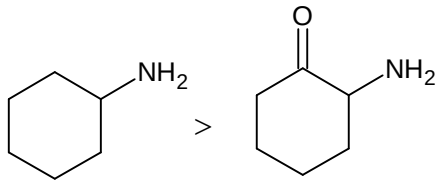
(B) Lysozyme (C) will migrate to the anode

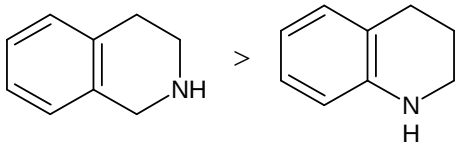
(C) Haemoglobin will not migrate

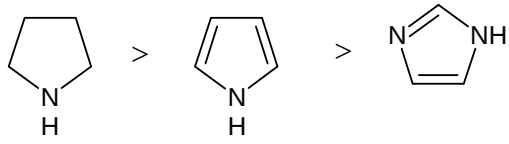
(D) at $pH = 7$, (A) and (C) would precipitate out while (B) would remain in solution.

12. Which of the following order for basic strength is/are correct?

(A) 

(B) 

(C) 

(D) 

13. Which of the following has face centered Bravais lattice?

(A) Hexagonal (B) Monoclinic

(C) cubic (D) orthorhombic

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. If four atoms of same radius are placed at the alternate corner of a cube touching each other, than the length of body diagonal of the cube is equal to $\sqrt{x} \times R$, where R is the radius of atom. Find the value of x?

2. The cost of electricity required to deposit 1 g Mg is Rs. 5.00. The cost of 30 g of Al to be deposited is Rs. X. Find the value of $\frac{x}{40}$?

3. A certain buffer solution contains equal concentration of X^- and HX. The K_b for X^- is 10^{-10}. Find the pH of this buffer solution?

4. Co forms dinuclear complex with a sigma bond within two Co atoms. Consider that metal carbonyls follows EAN rule. The complex can be written as $\mathrm{CO}_2(CO)_x$. Find the value of x ?

5. Diethyl $-\beta$, $\beta-$ dimethylglutaric acid $(ester)$ is condensed with diethyl oxalate in presence of sodium ethoxide and ethanol to form a major product (P), which on acidic hydrolysis followed by heating gives another product (Q). Calculate total number of C=O bonds in (P) and (Q) compounds?

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **5 multiple choice questions** numbered 1 to 5. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. Suppose f and g are functions having second derivatives f" and g" everywhere, if f(x) g(x) = 1 for all x and f ' (x) and g' are never zero, then $\frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)}$ equals

(A) $\frac{f'(x)}{f(x)}$ (B) $2\frac{f'(x)}{f(x)}$

(C) $-\frac{2g'(x)}{g(x)}$ (D) $-\frac{2f'(x)}{f(x)}$

2. Let a real valued differentiable function f(x) be strictly monotonic on [a, b], then

(A) $\int_a^b \left[(f(x))^2 - (f(b))^2\right]dx = \int_{f(a)}^{f(b)} 2x\left(b - f^{-1}(x)\right)dx$

(B) $\int_a^b \left[(f(x))^2 - (f(a))^2\right]dx = \int_{f(a)}^{f(b)} 2x\left(b - f^{-1}(x)\right)dx$

(C) $\int_a^b [f(x) - f(a)]^2\,dx = \int_{f(a)}^{f(b)} x\left(b - f^{-1}(x)\right)^2 dx$

(D) None of these

Space for rough work

3. Let A be the point of intersection of the curve $y = \log_{\frac{1}{3}}\left(x - \frac{1}{3}\right) + \log_3 \sqrt{9x^2 - 6x + 1}$ and the circle $x^2 + y^2 = 17$. B is also a point on the curve but lies inside the given circle such that its abscissa is an integer, then max {AB} is
(A) 2 units (B) 3 units
(C) 4 units (D) none of these

4. If S_n denotes the sum of first n terms of an AP and $a_n = \frac{S_{3n}}{S_{2n} - S_n}$, then $\underset{n\to\infty}{\text{Lt}} \sum_{r=1}^{2n} \frac{a_r}{n}$ is equal to
(A) 2 (B) 4
(C) 6 (D) 8

5. The possible value of a for which $e^{2x} - (a - 2)e^x + a < 0$ holds for at least one positive x is
(A) 2 (B) 3
(C) 4 (D) 7

Multiple Correct Choice Type

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

6. The values of x for which $x^2 - 4x > \cot^{-1} x$ holds true belong to
(A) $(-\infty, -1)$ (B) $[5, \infty)$
(C) $(-1, 5]$ (D) none of these

7. For a given parabola $y^2 = 4ax$, two variable chords *PQ* and *RS* at right angles are drawn through the fixed point $A(x_1, y_1)$ inside the parabola, making variable angles θ and α with *x*–axis. If r_1, r_2, r_3, r_4 are distances of *P*, *Q*, *R* and *S* from *A*, then the value of $\frac{1}{r_1 r_2} + \frac{1}{r_3 r_4}$
(A) independent of θ (B) independent of α
(C) depends upon both θ and α (D) is a constant

Space for rough work

8. From (1, 2), tangents are drawn to the curve $y^2 - 2x^3 - 4y + 8 = 0$. Then
(A) sum of x–coordinates of points of contact is zero
(B) sum of x–coordinates of points of contact is 4
(C) sum of y–coordinates of points of contact is zero
(D) sum of y–coordinates of points of contact is 4

9. If $\left|\int_a^b f(x)dx\right| = \int_a^b |f(x)|dx \ \ (a \neq b)$ also $f(x) \neq 0$ for any $x \in (a, b)$ and $g(x) = \int_0^x f(x)dx$, then $\int_a^b f(x)g(x)dx$ when a and b are positive
(A) cannot be positive
(B) cannot be negative
(C) cannot be equal to zero
(D) nothing can be said

10. If $\alpha, \beta, \gamma \in R$, then the determinant
$$\Delta = \begin{vmatrix} (e^{i\alpha} + e^{-i\alpha})^2 & (e^{i\alpha} - e^{-i\alpha})^2 & 4 \\ (e^{i\beta} + e^{-i\beta})^2 & (e^{i\beta} - e^{-i\beta})^2 & 4 \\ (e^{i\gamma} + e^{-i\gamma})^2 & (e^{i\gamma} - e^{-i\gamma})^2 & 4 \end{vmatrix} \text{ is}$$
(A) equal to 0
(B) equal to $e^{i(\alpha+\beta+\gamma)}$
(C) independent of α, β and γ
(D) dependent of α, β and

Space for rough work

11. If AB = A and BA = B, where A and B are square matrices, then
(A) $A^2 = A$ (B) $A^2 = B$
(C) $B^2 = B$ (D) $B^2 = A$

12. If a complex number satisfies the equation $|z| - \log_2(z + 2) = 0$ (Re z > 0), then
(A) $z = -\bar{z}$ (B) $|z| = 2$
(C) $\arg(z) = \frac{\pi}{2}$ (D) $\arg(z) = 0$

13. If the equation $x^3 + 2x^2 + x + k = 0$ has three real roots in which two roots are equal, then value of k is/are
(A) 0 (B) $k = \frac{27}{4}$
(C) $k = \frac{4}{27}$ (D) none of these

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Find number of positive integral solution of equation $\tan^{-1} x + \cos^{-1} \frac{y}{\sqrt{1-y^2}} = \sin^{-1} \frac{3}{\sqrt{10}}$

2. Tangent are drawn from any point on the hyperbola $x^2/9 - y^2/4 = 1$ to the circle $x^2 + y^2 = 9$ If the locus of the mid-point of the chord of contact is a $a\left(x^2 + y^2\right)^2 = bx^2 - cy^2$, then the value of $a^2 + b^2 + c^2 - 7870$ is equal to……..

Space for rough work

3. If $A=\begin{pmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{pmatrix}$ and $A^2 - 4A - 5I = O$ where I and O are the unit matrix and the null matrix of order 3 respectively. If $15A^{-1}=\lambda\begin{vmatrix}-3 & 2 & 2\\ 2 & -3 & 2\\ 2 & 2 & -3\end{vmatrix}$ then the find the value of λ.

4. The plane $2x-2y+z=3$ is rotated about the line where it cuts the xy plane by an acute angle α. If the new position of plane contains the point (3, 1, 1) then $9\cos\alpha$ equal to …….

5. If $0 < \alpha, \beta < 2\pi$, then the number of ordered pairs (α, β) satisfying $\sin^2(\alpha+\beta) - 2\sin\alpha\sin(\alpha+\beta) + \sin^2\alpha + \cos^2\beta = 0$ is

Space for rough work